Nonlinear Regression Analysis of Data Set 94

MTH 686 (Nonlinear Regression), IIT Kanpur, course project

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27 October 2026

1 Problem and summary

This is a revised and extended version (October 2026) of the six-page report submitted on 12 November 2025. It adds the explanation of why Model 1's two-exponential fit breaks down, the Osborne cross-check, working profile-likelihood intervals, bootstrap intervals, and the two extensions in §10–11.

Data set 94 consists of $n = 70$ observations $(t, y(t))$, $t = 1, \ldots, 70$, assumed to follow one of

$$\begin{aligned}
\text{Model 1:} \quad & y(t) = \alpha_0 + \alpha_1 e^{\beta_1 t} + \alpha_2 e^{\beta_2 t} + \varepsilon(t), \\
\text{Model 2:} \quad & y(t) = \frac{\alpha_0 + \alpha_1 t}{\beta_0 + \beta_1 t} + \varepsilon(t), \\
\text{Model 3:} \quad & y(t) = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3 + \beta_4 t^4 + \varepsilon(t),
\end{aligned}$$

with $\varepsilon(t) \overset{iid}{\sim} N(0, \sigma^2)$.

Summary. The data are best described by **Model 1**, but not in its two-exponential form. Model 1's least-squares estimator *does not exist* for this data set. The residual sum of squares decreases monotonically as the two rates merge ($\beta_2 - \beta_1 \to 0$) while $\alpha_1 = -\alpha_2 \to \infty$. Two independent methods confirm this: variable-projection Levenberg–Marquardt and Osborne's modified Prony algorithm. The infimum is attained by the *confluent* limit $\alpha_0 + (\alpha_1 + \gamma t)e^{\beta t}$. That limit in turn is not significantly better than a single exponential ($F = 2.2$, $p = 0.143$), so the selected model is

$$\hat{y}(t) = 0.761 + 3.730\, e^{-0.372\, t}, \qquad \hat{\sigma}^2 = 0.130.$$

Two technical points recur below:

- **Model 2 as stated is not identifiable.** Scaling $(\alpha_0, \alpha_1, \beta_0, \beta_1)$ by any $c \neq 0$ leaves the curve unchanged, so its Fisher information matrix is singular. We fix $\beta_0 = 1$.
- **All three models are partially linear**, which we exploit to reduce each non-convex search to one or two dimensions.

Two extensions follow the eight questions. §10 fits the *decaying wave* that Osborne's algorithm points to and checks whether the wave is real (it is not). §11 checks the whole method on simulated data where the right answer is known.

2 Least-squares estimators (Q1)

Least-squares estimates minimise $S(\theta) = \sum_t \{y_t - \mu(t;\theta)\}^2$. The table gives each model's starting value (from Section 3) and final estimate. Model *M1c* is the confluent limit of Model 1 and *M1s* its one-exponential sub-model ($\alpha_2 = 0$). Model 2 uses $\beta_0 = 1$, so its parameters are $(\alpha_0, \alpha_1, \beta_1)$, written `a0, a1, b1`.

The *decaying wave* M1w (§10) is listed with the fits for completeness.

Table 1: Starting values and least-squares estimates. Row M1 is where Levenberg–Marquardt stopped after 500 iterations, *not* a converged estimate (see §3.2).

model	parameter	start	estimate
M1	a0	0.771825	0.771798
M1	a1	-23.9346	-235.456
M1	a2	23.2988	235.08
M1	b1	-0.983932	-0.84136
M1	b2	-0.725976	-0.817094
M1c	a0	0.771861	0.7718
M1c	a1	-0.401866	-0.375135
M1c	g	5.74089	5.70699
M1c	b1	-0.830592	-0.829045
M1s	a0	0.761501	0.761474
M1s	a1	3.73123	3.73019
M1s	b1	-0.372224	-0.372068
M1w	a0	0.773199	0.7733
M1w	c	-0.0943936	0.469511
M1w	d	22.2956	15.0727
M1w	rho	-0.772287	-0.706738
M1w	w	0.231633	0.287142
M2	a0	7.70283	7.70127
M2	a1	1.01124	1.0109
M2	b1	1.62378	1.62326
M3	b0	2.69422	2.69422
M3	b1	-0.260131	-0.260131
M3	b2	0.0116652	0.0116652
M3	b3	-0.000209401	-0.000209401
M3	b4	1.29233e-06	1.29233e-06

3 How the estimators were found, and initial values (Q2)

3.1 Variable projection

Each model is linear in its amplitudes once the nonlinear parameters are fixed. For Model 1, given (β_1, β_2), the amplitudes $(\alpha_0, \alpha_1, \alpha_2)$ are the OLS coefficients of y on $[1, e^{\beta_1 t}, e^{\beta_2 t}]$. Substituting them back gives a *profiled* criterion in the rates alone (Golub–Pereyra variable projection):

$$S^*(\beta_1, \beta_2) = \min_{\alpha} S(\alpha, \beta) = y^\top (I - P_{X(\beta)})\, y.$$

This turns a five-dimensional non-convex problem into a two-dimensional one that can be scanned exhaustively. We scanned 80 log-spaced rates in $[-3, -10^{-3}]$ with $\beta_1 < \beta_2$; Models M1c and M1s are profiled over one rate. Model 2 (with $\beta_0 = 1$) is linear in (α_0, α_1) given β_1, so we profile over a 450-point grid of $\beta_1 > -1/70$, which keeps the denominator positive. The best grid point seeds a joint **Levenberg–Marquardt** refinement of all parameters with analytic Jacobians. Model 3 is linear and solved in closed form by QR.

Table 2: Convergence. The gradient norm is $\|J^\top r\|$ at the final iterate.

model	iterations	converged	grad_norm	RSS
M1	500	FALSE	0.00283752	8.42818
M1c	4	TRUE	1.22749e-08	8.42817
M1s	4	TRUE	6.37449e-07	8.70864
M1w	17	TRUE	3.36786e-07	8.42625
M2	4	TRUE	1.71553e-07	9.10391
M3	0	TRUE	1.39896e-06	10.9099

M1c, M1s and M2 converge in at most 4 iterations (the wave M1w of §10 in 17) to stationary points with gradient

norm below 10^{-6}, and their estimates move away from the grid starts. The grid only seeds the optimiser; it does not determine the answer.

3.2 Model 1 has no interior least-squares estimate

Started from the best grid point, Levenberg–Marquardt on Model 1 never converges. It drifts steadily toward the boundary $\beta_1 = \beta_2$, with the amplitudes diverging in opposite directions while the RSS keeps decreasing:

Table 3: Levenberg–Marquardt trajectory for Model 1.

iterations	a1	a2	b1	b2	rate_gap	RSS
10	-27.46	26.98	-0.9512	-0.7352	0.2161	8.429
50	-35.47	35.03	-0.9198	-0.755	0.1648	8.428
200	-83.05	82.66	-0.8651	-0.796	0.06909	8.428
1000	-527.3	526.9	-0.8345	-0.8237	0.01083	8.428
5000	-1224	1224	-0.8314	-0.8267	0.004662	8.428

As $\beta_2 \to \beta_1$ with $\alpha_1(\beta_2 - \beta_1) \to \gamma$, the curve converges to $\alpha_0 + (\alpha_1 + \gamma t)e^{\beta t}$. Fitted directly, this model (M1c) attains RSS = 8.428172, the infimum the Model 1 path approaches. R's built-in `nls(algorithm = "plinear")` agrees: it fails (step factor 0.000488281 reduced below minFactor of 0.000976562). The profiled surface in Figure 1 shows the valley floor running into the diagonal.

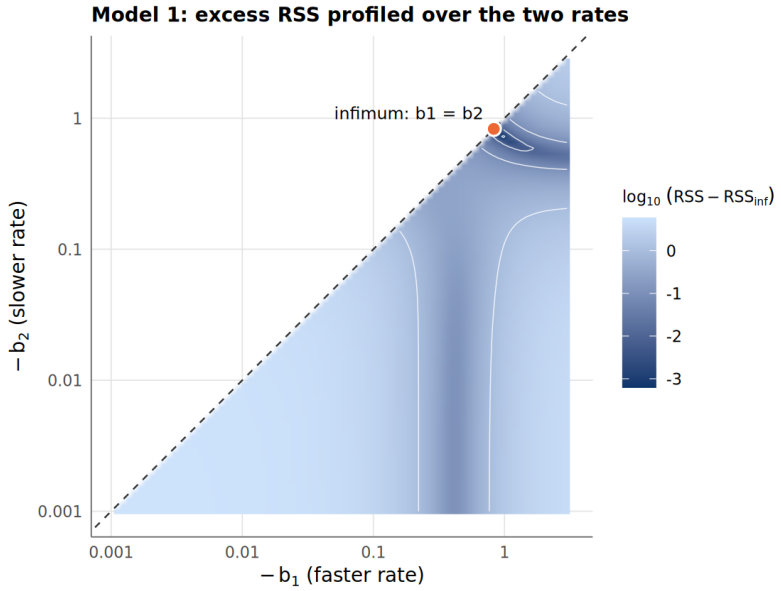


Figure 1: Profiled excess RSS of Model 1 over its two rates. The minimum lies on the excluded diagonal $\beta_1 = \beta_2$.

3.3 Independent check: Osborne's modified Prony algorithm

For equally spaced t, $\mu_t = \alpha_0 + \alpha_1 x_1^t + \alpha_2 x_2^t$ (with $x_k = e^{\beta_k}$) satisfies a linear recurrence whose characteristic polynomial $g(x)$ has roots $\{1, x_1, x_2\}$. We impose the root at 1 exactly by writing $g(x) = (x-1)h(x)$ with h quadratic. The least-squares problem then becomes minimising $R(h) = y^\top P_{G(h)}\, y$, and Osborne's iteration solves $B(h)h = 0$ through repeated smallest-eigenvalue eigenvectors. (Deriving B gives a *minus* sign on its middle term, where the course notes print a plus. $B(h)h$ is the gradient of R, and at the 11 starting points it matches a finite-difference gradient to a relative error of 3.8×10^{-6}.)

The iteration only finds *a* stationary point, so it was run from the classical Prony start and 10 rate pairs:

Table 4: Osborne runs. R is the RSS of the best fit in the recurrence class. A run counts as converged when h moves by less than 10^{-8} in one step.

start	R	iterations	converged	roots	real_positive
prony	20.3267	23	TRUE	-0.9786+0.2938i, -0.9786-0.2938i	FALSE
b=(-0.05,-0.2)	8.52714	15	TRUE	0.6644-2.019e-28i, 1.0166+2.019e-28i	TRUE
b=(-0.05,-0.5)	8.52714	17	TRUE	0.6644+5.049e-29i, 1.0166-5.049e-29i	TRUE
b=(-0.05,-1)	8.52714	17	TRUE	0.6644-5.049e-29i, 1.0166+5.049e-29i	TRUE
b=(-0.05,-2)	8.52714	17	TRUE	0.6644+0i, 1.0166+0i	TRUE
b=(-0.2,-0.5)	8.52714	21	TRUE	0.6644-1.515e-28i, 1.0166+1.515e-28i	TRUE
b=(-0.2,-1)	8.52714	19	TRUE	0.6644-5.049e-29i, 1.0166+5.049e-29i	TRUE
b=(-0.2,-2)	8.52714	19	TRUE	0.6644-5.049e-29i, 1.0166+5.049e-29i	TRUE
b=(-0.5,-1)	8.52714	20	TRUE	0.6644-1.01e-28i, 1.0166+1.01e-28i	TRUE
b=(-0.5,-2)	8.42625	28	TRUE	0.4731+0.1397i, 0.4731-0.1397i	FALSE
b=(-1,-2)	20.3267	159	TRUE	0.9487+0.6924i, 0.9487-0.6924i	FALSE

The other starts stop at worse stationary points (larger R in Table 4). Most of these have one real root slightly above 1, a slowly *growing* exponential. Only the best stationary point is used below.

The best stationary point, $R = 8.426248$, has **complex-conjugate roots** 0.4731+0.1397i, 0.4731-0.1397i. That is a damped oscillation ($|x| = 0.4933$, frequency 0.2871 rad), which no real-rate Model 1 can represent. The minimum over admissible h (real, positive roots) therefore lies on the boundary of that region, where the discriminant is zero: a double root. This confirms the variable-projection result. In Osborne's h-coordinates the double root is a regular point, which explains why that parametrisation is better behaved than (α, β) here.

4 The best fitted model (Q3)

We compare fits by information criteria. Here $k = p + 1$ counts σ, $\log L$ is the Gaussian log-likelihood at the MLE, and Model 1 is represented by its closure M1c and its sub-model M1s.

Table 5: Model comparison, sorted by AIC.

model	p	RSS	sigma2	AIC	AICc	BIC
M1c	4	8.428	0.1277	60.47	61.4	71.71
M1s	3	8.709	0.13	60.76	61.37	69.75
M1w	5	8.426	0.1296	62.45	63.78	75.94
M2	3	9.104	0.1359	63.87	64.48	72.86
M3	5	10.91	0.1678	80.53	81.87	94.02

Relative to the best Model 1 fit, the rational model is worse by 3.4 in AIC and the polynomial by 20, so the exponential family clearly wins. Within it, M1s is nested in M1c ($\gamma = 0$, a regular hypothesis because β stays identified), so an extra-sum-of-squares F test applies:

Table 6: Nested test of M1s inside M1c.

test	df1	df2	F	p_value	LR	p_value_LR
M1s (one exponential) vs M1c (confluent)	1	66	2.196	0.1431	2.292	0.1301

With $p = 0.143$ the second term is not supported. BIC also prefers M1s, and AIC differs by only 0.3. **Best model: Model 1 with a single exponential term**, $\hat{y}(t) = 0.761 + 3.730\, e^{-0.372\, t}$.

5 Estimate of σ^2 (Q4)

$\hat{\sigma}^2 = \mathrm{RSS}/(n - p)$ is given in the σ^2 column of Table 5. For the selected model, $\hat{\sigma}^2 = 0.130$ ($\hat{\sigma} = 0.361$).

6 Confidence intervals from the Fisher information (Q5)

Under normal errors, $I(\theta) = J^\top J/\sigma^2$ with $J = \partial\mu/\partial\theta$ at $\hat\theta$, and the Wald interval is $\hat\theta_j \pm t_{n-p,\,0.975}\sqrt{\hat\sigma^2[(J^\top J)^{-1}]_{jj}}$. Wald intervals assume the model is close to linear in θ. We check that assumption against **profile-likelihood** intervals $\{c : S_{(j)}(c) \le S(\hat\theta)(1 + F_{1,n-p;0.95}/(n-p))\}$ and against 2000-replicate **residual-bootstrap** percentile intervals.

Table 7: 95% confidence intervals. NA means the profile does not cross the cutoff within 30 standard errors, so the interval is unbounded.

model	param	estimate	SE	Wald lo	Wald hi	profile lo	profile hi	boot lo	boot hi
M1c	a0	0.7718	0.04525	0.6814	0.8622	0.6812	0.8619	0.6775	0.8621
M1c	a1	-0.3751	4.323	-9.007	8.257	-19.55	4.676	-19.9	4.062
M1c	g	5.707	5.147	-4.569	15.98	-0.4981	28.05	7.611e-07	28.42
M1c	b1	-0.829	0.2265	-1.281	-0.3768	-1.363	-0.1986	-1.372	-0.3292
M1s	a0	0.7615	0.04689	0.6679	0.8551	0.6677	0.8542	0.6664	0.8539
M1s	a1	3.73	0.6697	2.394	5.067	2.588	5.239	2.644	5.453
M1s	b1	-0.3721	0.08082	-0.5334	-0.2107	-0.5587	-0.2413	-0.5898	-0.236
M1w	a0	0.7733	0.04655	0.6803	0.8663	0.6807	0.8671	0.6906	0.8816
M1w	c	0.4695	6.117	-11.75	12.69	-19.8	4.69	-6.347	4.252
M1w	d	15.07	68.05	-120.8	151	NA	NA	-2.567	41.32
M1w	rho	-0.7067	0.8196	-2.344	0.9301	-1.367	-0.1976	-0.9625	-0.1598
M1w	w	0.2871	0.733	-1.177	1.751	-0.66	0.66	-2.058e-08	0.6483
M2	a0	7.701	4.619	-1.519	16.92	3.859	NA	3.785	5.895e+10
M2	a1	1.011	1.004	-0.9922	3.014	0.2604	NA	0.2021	1.322e+10
M2	b1	1.623	1.493	-1.357	4.604	0.4893	NA	0.4096	2.017e+10
M3	b0	2.694	0.2674	2.16	3.228	2.16	3.228	2.178	3.25
M3	b1	-0.2601	0.0514	-0.3628	-0.1575	-0.3628	-0.1575	-0.3646	-0.1607
M3	b2	0.01167	0.002915	0.005845	0.01749	0.005845	0.01749	0.005812	0.01748
M3	b3	-0.0002094	6.147e-05	-0.0003322	-8.664e-05	-0.0003322	-8.664e-05	-0.0003313	-8.39e-05
M3	b4	1.292e-06	4.296e-07	4.344e-07	2.15e-06	4.344e-07	2.15e-06	4.245e-07	2.135e-06

Table 8: Identifiability. Condition number of the column-scaled $J^\top J$, and the largest absolute correlation between estimates.

model	condition_number	max_abs_correlation
M1	6.49e+12	1
M1c	392	0.976
M1s	12.1	0.808
M1w	9.12e+03	0.992
M2	1.37e+03	0.996
M3	2.31e+05	0.992

- **M1s, the selected model:** all three intervals agree closely, and the model is well conditioned ($\kappa \approx 12$). Its Wald intervals can be trusted.
- **M3:** linear, so Wald and profile intervals coincide exactly.
- **M2:** although $\beta_0 = 1$ makes it identifiable, it is *weakly* identified. The profile intervals for all three parameters are unbounded above and the bootstrap upper limits explode, because the data mainly determine the ratio α_1/β_1 (the asymptote). Its symmetric Wald intervals hide this.
- **M1, as originally parametrised:** $J^\top J$ is numerically singular ($\kappa \approx 6.5 \times 10^{12}$). This is why a naive fit reports huge standard errors: where Levenberg–Marquardt stopped, about 6.8×10^5 for the amplitudes and 36 for the rates, and they keep growing the longer it runs.

7 Residuals (Q6)

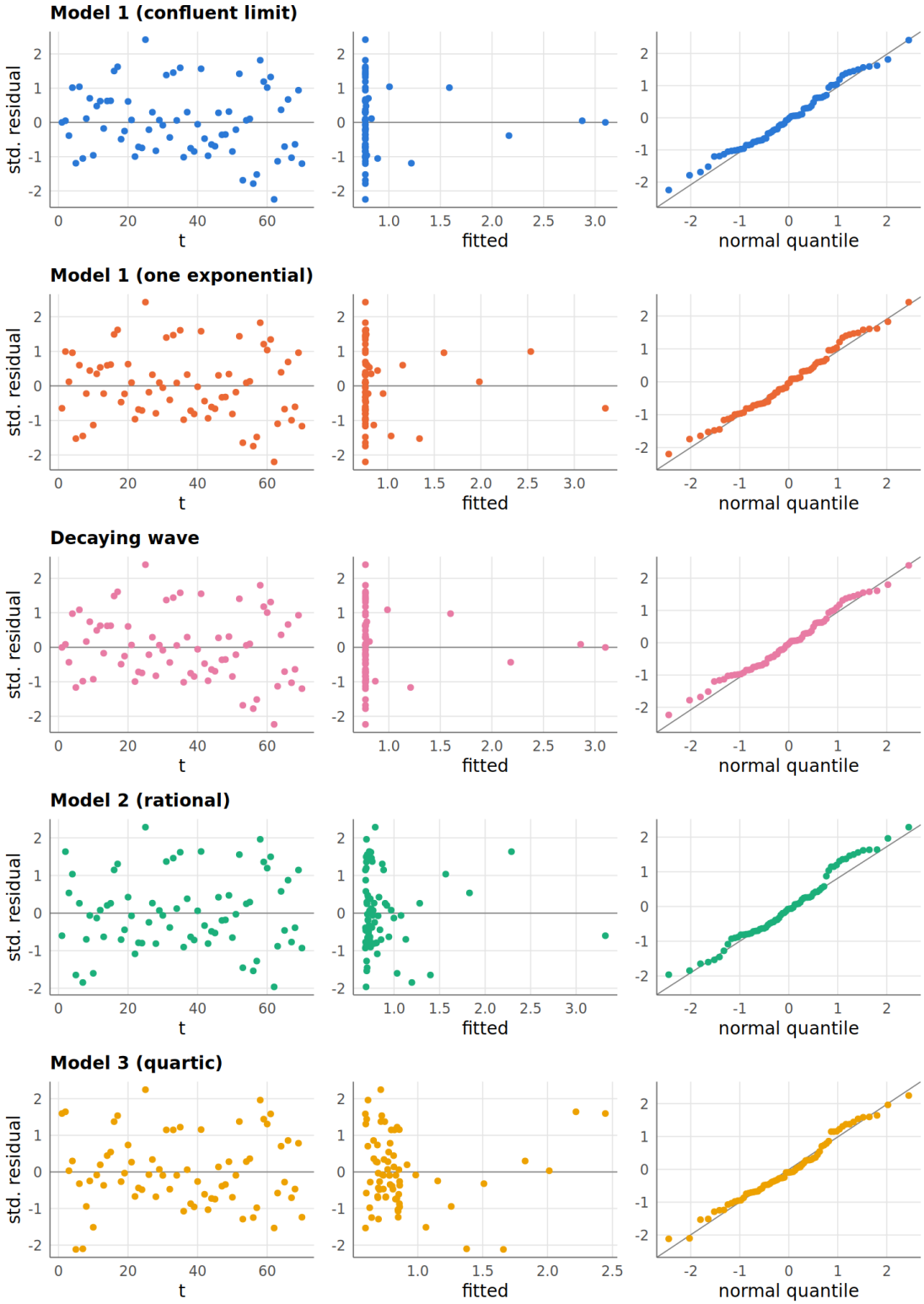


Figure 2: Standardised residuals against t and fitted values, and normal Q–Q plots.

None of the models shows trend or funnelling against t or the fitted values. The Q–Q plots are close to linear.

8 Normality of the errors (Q7)

Table 9: Shapiro–Wilk (SW), Anderson–Darling (AD) and Jarque–Bera (JB) normality tests, and the Wald–Wolfowitz runs test on residual signs.

model	SW W	SW p	AD p	JB p	runs	E[runs]	runs p
M1c	0.985	0.563	0.323	0.558	44	36	0.0541
M1s	0.987	0.68	0.511	0.556	43	36	0.0903
M1w	0.985	0.554	0.312	0.558	45	36	0.0296
M2	0.975	0.164	0.0878	0.42	41	35.7	0.202
M3	0.98	0.338	0.161	0.584	40	35.5	0.277

For the selected model none of the three tests rejects normality at the 5% level (SW $p = 0.68$). The runs test shows no systematic lack of fit ($p = 0.09$).

9 Observed data and fitted curves (Q8)

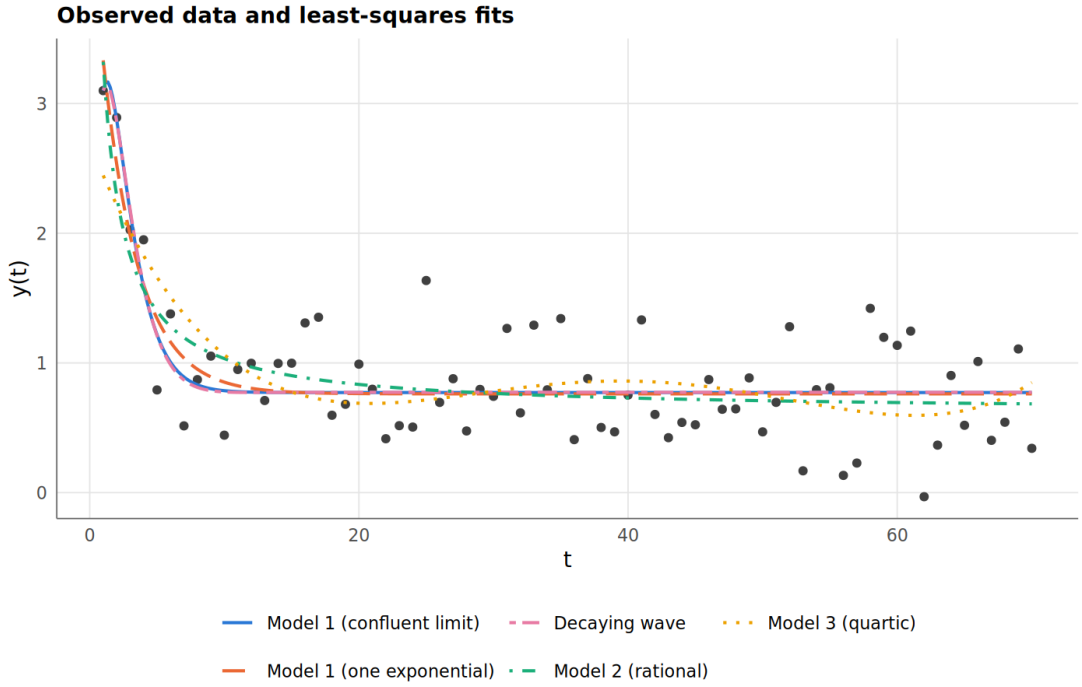


Figure 3: Data with all least-squares fits. The rational and quartic fits bend away from the data near $t = 1$ and in the tail.

10 Extension: is the wave real?

Osborne's algorithm (§3.3) says the best curve of Model 1's general form is a *decaying wave* rather than a sum of exponentials:

$$y(t) = a_0 + e^{\rho t}\,(c\cos\omega t + d\sin\omega t) + \varepsilon(t).$$

We fit it the same way as the other models: a grid over (ρ, ω), with (a_0, c, d) solved by least squares at each grid point, then Levenberg–Marquardt. It reaches RSS = 8.426248, exactly Osborne's value, which is a third independent check of §3. The wave is heavily damped: its amplitude shrinks by a factor of 0.493 per time step, so although its period is 21.9, it has died out long before completing one cycle.

Compared with the one-exponential model it lowers the RSS by only 0.282. To judge whether that gain is meaningful, we made 1000 fake data sets from the fitted one-exponential model, which by construction has no wave, and fitted both models to each. On 22% of them the wave improved the RSS by at least as much as on

the real data (Figure 4). That is the p-value of the test. The improvement on the real data is what noise alone produces, so **the wave is not real**, and the simple model stands.

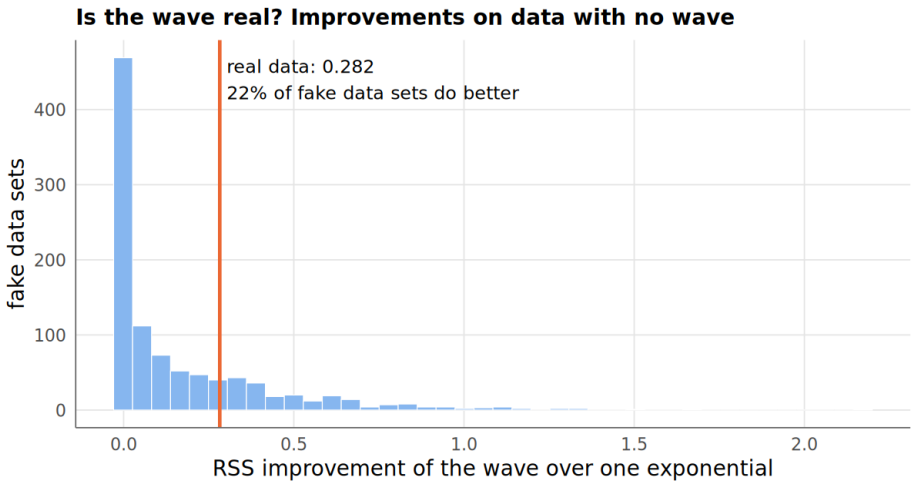


Figure 4: RSS improvement of the wave model on fake data sets with no wave. The orange line marks the real data.

11 Extension: checking the method on simulated data

Does the method find the right answer when we know what it is? For each of the three fitted models (one exponential, rational, quartic) we generated 1000 fake data sets with the same t values and noise level, and ran the full pipeline on each one.

Does it pick the right model? Rows give the model the data came from; columns the model chosen by the rule in §4.

Table 10: Model chosen, out of 1000 fake data sets per true model.

truth	chose M1s	chose M1c	chose M2	chose M3
true M1s	761	79	155	5
true M2	243	1	750	6
true M3	342	58	101	499

The right model is chosen 76% of the time for the one-exponential truth and 75% for the rational one. The quartic is recovered only 50% of the time: with this much noise, a quartic curve is often fitted just as well by a decaying exponential with fewer parameters. With 70 noisy points the three families are not always distinguishable, and the selection in §4 should be read with that in mind.

Do the 95% intervals really cover 95%? For each fake data set we computed the Wald intervals of the true model and counted how often they contained the true parameter values.

Table 11: Coverage of 95% Wald intervals. "Undefined" counts fits whose standard errors could not be computed.

truth	parameters	coverage	undefined
true M1s	a0, a1, b1	95.7%, 95.8%, 94.6%	0
true M2	a0, a1, b1	86.8%, 85.0%, 85.8%	42
true M3	b0, b1, b2, b3, b4	94.2%, 94.3%, 93.7%, 93.3%, 93.2%	0

For the quartic, a linear model, the coverage must be exactly 95%. Here it is 93–94%: with 1000 data sets the estimated coverage itself varies by about ±1.4 points, and the five parameters move together. A separate run of 10,000 data sets for the quartic alone gives 94.7–95.3%, which confirms that the simulation and the

interval formula are correct. The selected one-exponential model is also close to 95%, so its intervals in §6 can be trusted. The rational model falls short, as expected from its weak identifiability (§6).

How often does the two-exponential Model 1 break down?

Table 12: Share of fake data sets on which the two-exponential fit has no proper solution (the two rates merge, as in §3.2).

truth	breaks down	fake data sets
true M1s	61%	1000
true M2	33%	1000
true M3	82%	1000

Even when the data truly come from a single exponential, the two-exponential fit breaks down on 61% of data sets. What happened with data set 94 is therefore typical, not bad luck: with one decay present and this much noise, a second exponential term usually has nothing to attach to.

Appendix: reproducing the results

All numbers, tables and figures come from `Rscript run_analysis.R`; the simulation results (§10–11) come from `Rscript simulate.R`. Source: `R/models.R` (mean functions and Jacobians), `R/fit.R` (variable projection and Levenberg–Marquardt), `R/osborne.R` (modified Prony), `R/inference.R` (information criteria, Wald, profile and bootstrap intervals), `R/diagnostics.R` (tests and figures), `R/simulation.R` (simulation checks).